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Intro To Cyber Security Lab - CY104L

Project Report

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RAT (Remote Access Trojans) Attacks By AhMyth

What are Remote Access Trojans?

Remote access trojans are the type of malware that allow an attacker to remotely control a victim's mobile phone or computer without their permission or their knowledge. RATs are usually attached to legitimate-looking apps or files, which once installed by the victim, create a backdoor access for him. Phishing emails, pirated or cracked software, malicious links or fake downloads are common ways RATs are used by attackers.

RATs are very dangerous because they grant full access of your device to an attacker without you even having a clue of it.

AhMyth

Purpose:

AhMyth is an open-source android-based remote access trojan tool that demonstrates how an attacker uses remote access trojans to remotely control and spy on android devices.

From an attacker's perspective, AhMyth's core purpose is unauthorized surveillance and data theft, whereas from a defense perspective, it is used to analyze mobile malware behavior.

Key Features:

The key features of AhMyth are:

1. Stealth operation: It will operate in the background, and the user will not be notified.
2. Client-server architecture: Enables the execution of remote commands.
3. Data exfiltration: Transmits information gathered to the attacker.
4. Permission abuse: Abuses permissions that Android provides to access sensitive data.
5. Persistence: Remains active after restarts.

Capabilities:

AhMyth can execute several malicious activities on an infected android device, such as:

1. Extract information regarding the device (model, version of operating system, network information).
2. Monitor the device's location.
3. Record audio through the microphone.

4. Capturing photographs with the device's cameras, both front and back.
5. Viewing stored files in the device.
6. Accessing Call logs and SMS messages.
7. Running commands remotely.

These features make AhMyth especially threatening as it accesses personal mobile information, which frequently carries such personal messages as conversations, bank details, credentials, and live location.

Configuration Details

The whole experimentation of AhMyth was conducted in a safe environment on virtual machines.

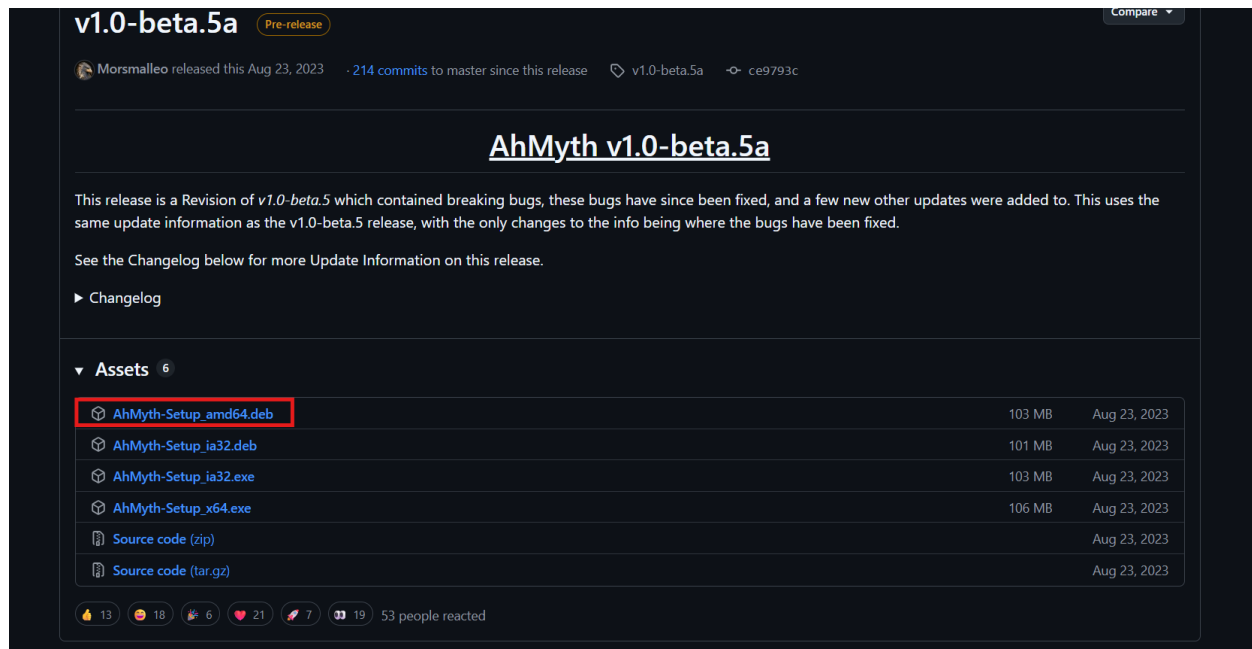
Virtual Machine Requirements:

Firstly, to set up the android virtual machine, we used Memu play, a virtual android environment. It was simply downloaded through; <https://www.memuplay.com/>. After completion of download, we set up a 64-bit, android version 9, virtual machine on it.

For the virtual machine to run AhMyth, we used Kali Linux installed on VMware.

Installation of AhMyth on Linux:

In Kali Linux, we downloaded AhMyth through GitHub, <https://github.com/Morsmalleo/AhMyth>. On this webpage we navigated to the binary setup section and selected the binary setup for Kali Linux, where we found the compatible version and the prerequisites and commands to be run on the terminal. In our case, we downloaded the latest 64-bit version.



Setup and Version Issues:

A common problem that occurs during AhMyth installation, which was also mentioned at the download webpage, is that it is compatible with java-11, but most new systems have java-21 version by default. Hence, we had to download and switch to the java-11 version.

We use the following command to install java-11:

```
sudo apt install openjdk-11-jdk.
```

Then we use the following command and select the java-11 version as default:

```
sudo update-alternatives --config java.
```

Following is the entire demonstration:

```
linux@kali: ~  
$ java --version  
openjdk 21.0.10-ea 2026-01-20  
OpenJDK Runtime Environment (build 21.0.10-ea+4-Debian-1)  
OpenJDK 64-Bit Server VM (build 21.0.10-ea+4-Debian-1, mixed mode, sharing)  
  
linux@kali: ~  
$ sudo apt install openjdk-11-jdk  
[sudo] password for linux:  
openjdk-11-jdk is already the newest version (11.0.25-9ea-1).  
The following packages were automatically installed and are no longer required:  
  amass-common libgeos3.14.0 libmongoc-1.0-0t64 libsqlcipher1 libx264-164 python3-kismetcapturebtgoiger python3-klrd tini  
  gir1.2-girepository-2.0 libgirepository-1.0-1 libnet1 libtheoradec1 libxml2 python3-kismetcapturefreaklabszigbee python3-klutls  
  libarmadillo4 libpgmutil164 libobjc-14-dev libtheoraenc1 libyelp0 python3-kismetcapturertl43 python3-klut python3-zombie-imp  
  libbluray2 libpgmpep6t64 libplacebo349 libbdfread0 linux-image-6.12.38+kali1-amd64 python3-kismetcapturertldsb python3-kismetcapturertlamr  
  libbson-1.0-0t64 libinstpatch-1.0-2 libportmidi0 libwirefreshk18 python3-bluepy python3-kismetcapturertlamr python3-multipart samba-ad-dc  
  libdisplay-info2 libjs-jquery-ui libradare2-5.0.0t64 libwiretap15 python3-click-plugins python3-multipart samba-ad-provision samba-dsdb-modules  
  libgdal7 libjs-underscore libravie0.7 libwautl16 python3-gpg python3-pysmi  
Use 'sudo apt autoremove' to remove them.  
  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
  
linux@kali: ~  
$ sudo update-alternatives --config java  
There are 2 choices for the alternative java (providing /usr/bin/java).  
  
  Selection Path Priority Status  
  * 0 /usr/lib/jvm/java-21-openjdk-amd64/bin/java 2111 auto mode  
    1 /usr/lib/jvm/java-11-openjdk-amd64/bin/java 1111 manual mode  
    2 /usr/lib/jvm/java-21-openjdk-amd64/bin/java 2111 manual mode  
  
Press center to keep the current choice[*], or type selection number: 1  
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/java to provide /usr/bin/java (java) in manual mode  
  
linux@kali: ~  
$ java --version  
openjdk 11.0.25-ea 2024-10-15  
OpenJDK Runtime Environment (build 11.0.25-ea+5-post-Debian-1)  
OpenJDK 64-Bit Server VM (build 11.0.25-ea+5-post-Debian-1, mixed mode, sharing)  
  
linux@kali: ~  
$
```

Then, we navigate to downloads and install the AhMyth file we downloaded, after which it becomes a runnable application on our virtual machine.

Lab Environment Description:

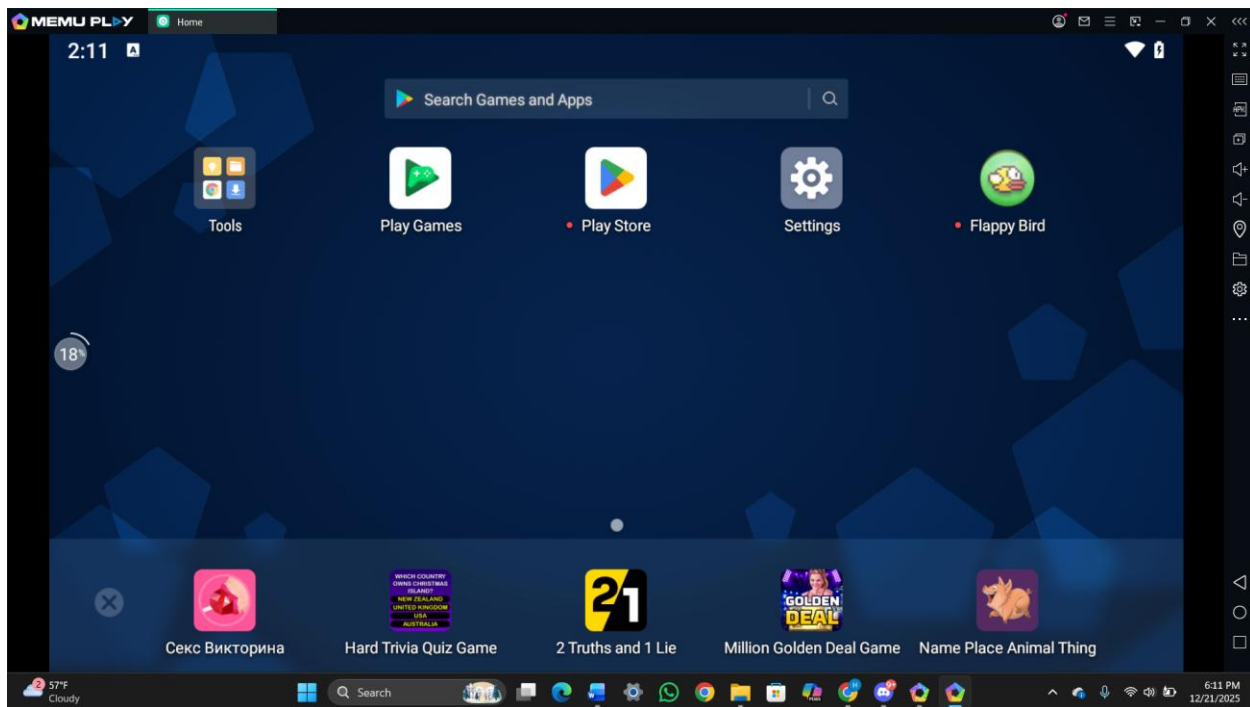
For the simulation of our tool, we used two approaches:

1. Running AhMyth on Virtual machines.
2. Running AhMyth on Windows and a dummy phone.

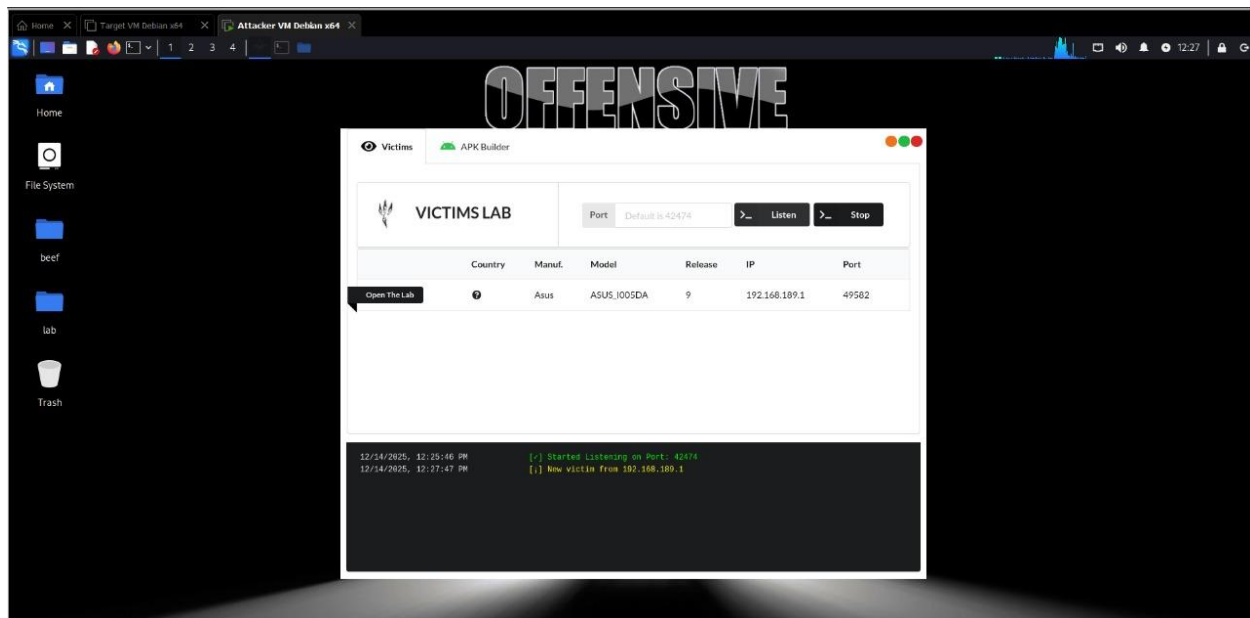
We will explain both of them briefly:

1. Running AhMyth on Virtual Machines:

To run AhMyth virtually, firstly, we set up the android virtual machine. we used Memu play, a virtual android environment. It was simply downloaded through; <https://www.memuplay.com/>. After completion of download, we set up a 64-bit, android version 9, virtual machine on it. It worked completely like a phone, and we could download all apks from our library on the phone.

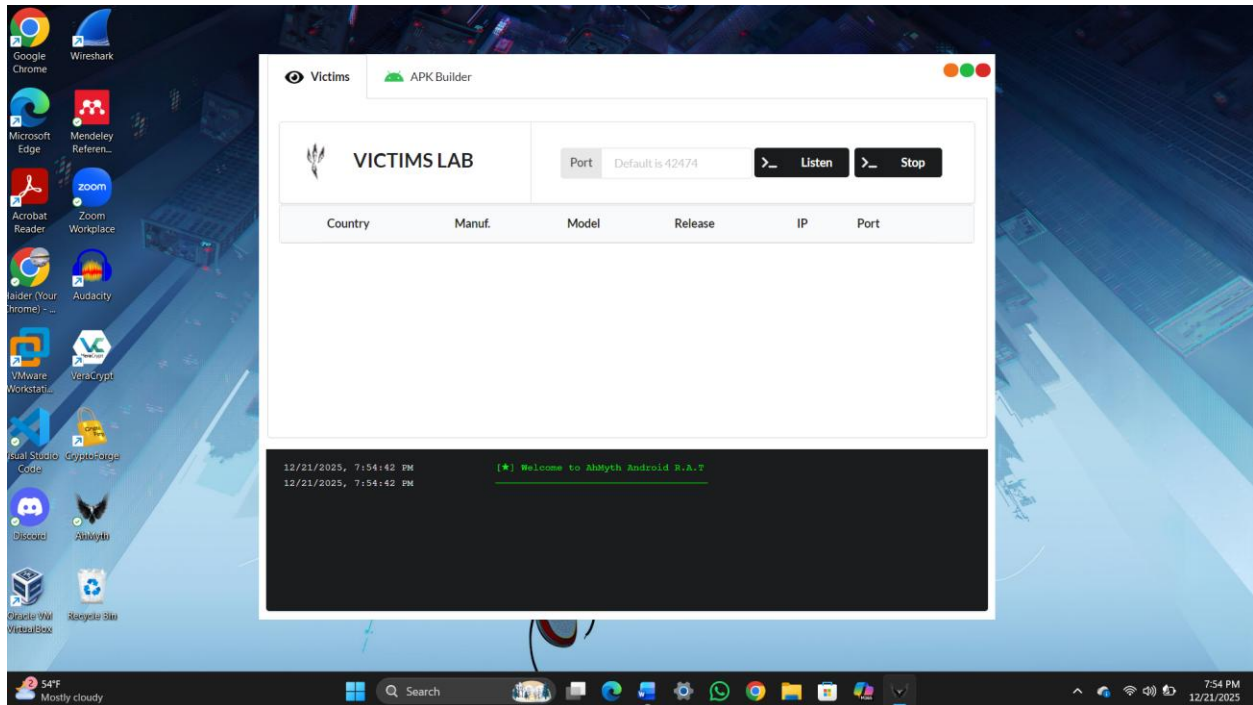


To setup Linux virtually, we went to <https://www.kali.org/get-kali/#kali-virtual-machines> and downloaded VMware, and Linux was downloaded via <https://www.kali.org/get-kali/#kali-installer-images> and run on VMWare. Then, after completion of all setup steps, we ran the command `sudo apt update && sudo apt upgrade -y` to install all latest applications and requirements, then we followed above mentioned ways to download AhMyth.



2. Running AhMyth on Windows and a Dummy phone:

The second secure approach to test AhMyth that we used was to download AhMyth on Windows and use it via a Dummy Phone. To download it on Windows, we simply applied the above mentioned downloading methods on a Windows browser.



The dummy phone that we used was a really old phone, a Huawei Y6 Pro which was in no one's use.

Simulation Of AhMyth

AhMyth's interface is very easy to navigate. To use remote access trojans by AhMyth, there are two ways:

1. Let AhMyth create an apk file with an attached malware.
2. Provide AhMyth your own apk file to attach a malware to it.

We used the 2nd method; however, We will explain both ways briefly.

1. Let AhMyth create an apk file with an attached malware.

The most easiest way to use AhMyth is to let AhMyth itself create the apk. In this method, AhMyth won't require an apk file from the user. After entering the IP Address of the device, AhMyth itself will make the file and save it to a directory on the virtual machine. That apk file is then used by the attacker and sent to people, who if download the apk, will appear as

victims on AhMyth's interface, and the attacker can then implement all his treacherous activities mentioned above on the victim's device.

2. Provide AhMyth your own apk file to attach a malware to it.

This method requires a user to have his/her own apk file to bind the malware to. After entering the user device's IP address, which we can find by running `ip` a command on the terminal, AhMyth will have an option to bind an apk file to the malware. There it will provide two further options of activity method and boot method. Their details are as follows:

1. Boot Method:

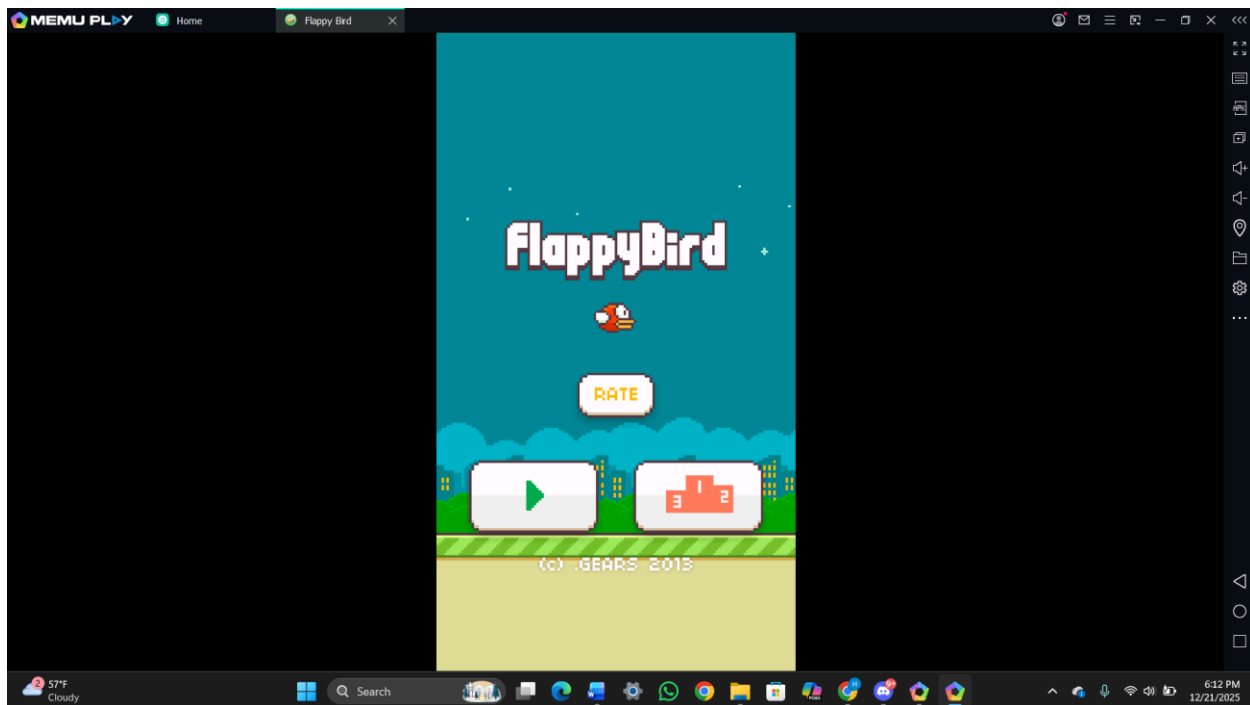
1. Runs automatically when the phone is switched on or restarted.
2. Does not need the user to open any app.
3. Works silently in the background.
4. Provides strong persistence and is hard to notice.

2. Activity Method:

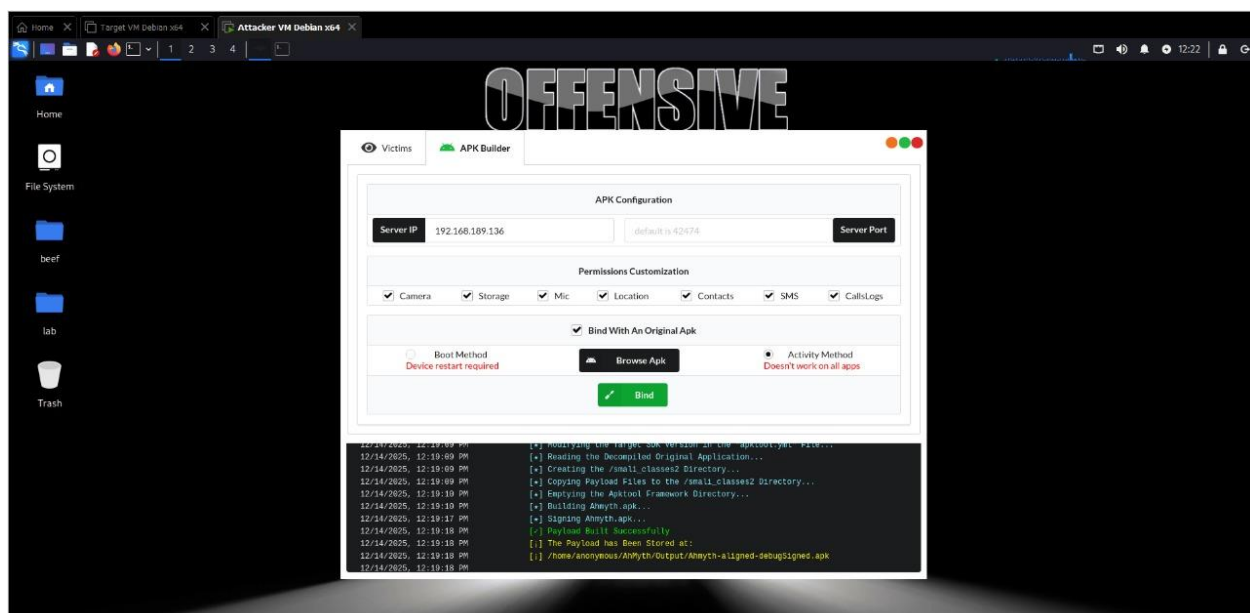
- Runs only when the user opens the app.
- Requires user interaction to start.
- Is more visible to the user.
- Stops when the app is closed.

Binding Malware to the APK:

After choosing any one of the methods according to the requirements of the attack, binding of own apk file is required. In our case, we searched the internet for an apk to bind the virus with. We found an game called Flappy Bird at <https://www.apkmirror.com/apk/gears-studios/flappy-bird/flappy-bird-1-3-release/flappy-bird-1-3-android-apk-download/> . This app was just a regular game. The following is its interface (captured on virtual android device).



After downloading the apk file, we provided it to AhMyth. When we provide an apk file to AhMyth, AhMyth uses its own methods to bind the malware to the apk file. Firstly, it decompiles the apk file, then it finds the correct location in it coding and inserts the malware to it. Then it reforms the apk file and saves it to a directory on the device.

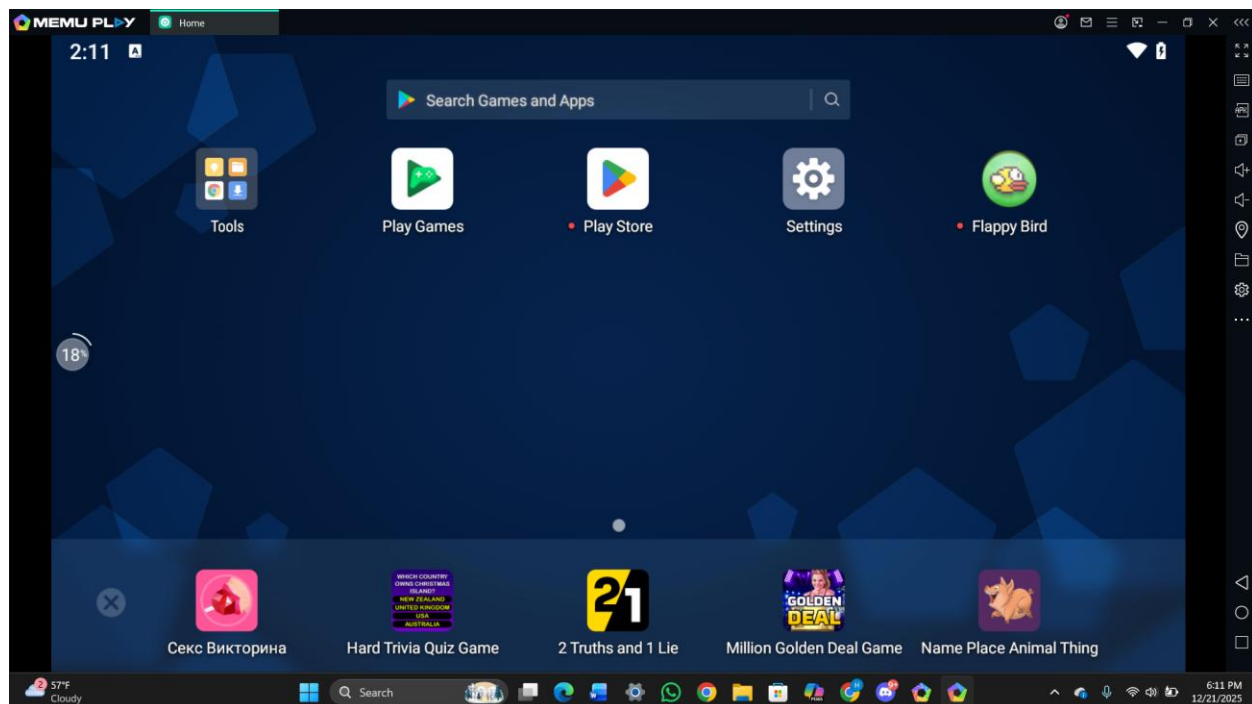


Execution of Attack:

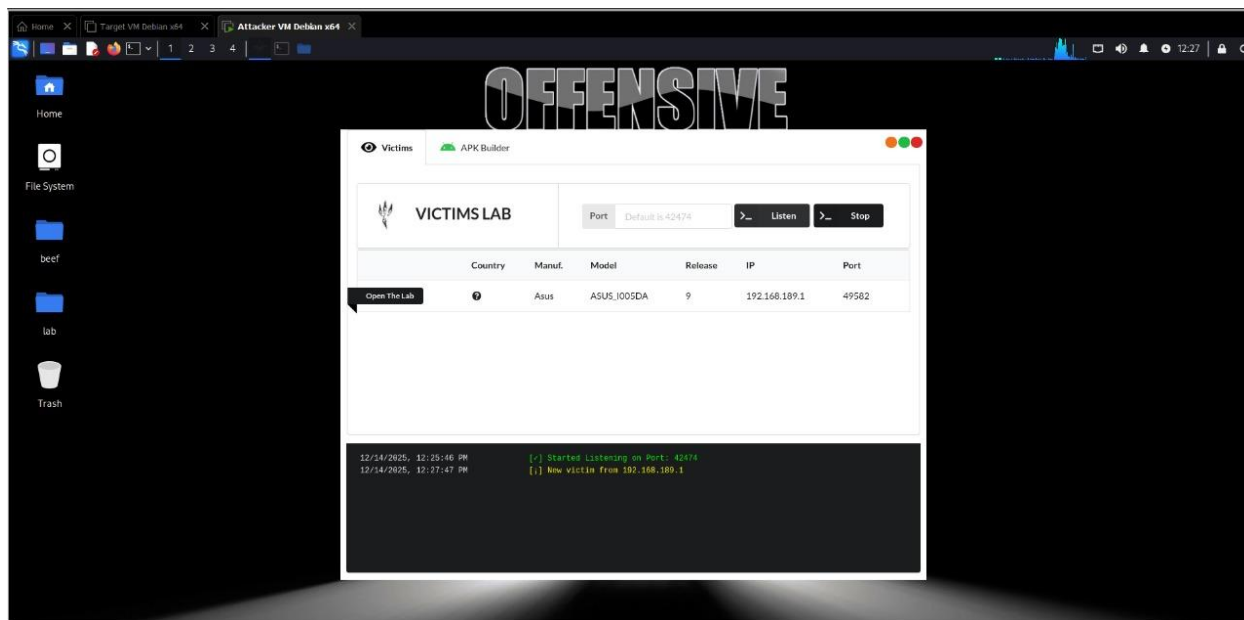
After the creation of the malware infused apk file, the attacker may use different ways to promote the application. Once that application is installed and opened by the victim,

AhMyth portrays its connection on its interface. From there onwards, if the app was made via boot method and is running in the background, the attacker may access anything in the phone, anytime, as long as the app is installed. If the app was created via activity method, the attacker will be able to access the information only when the user launches the app.

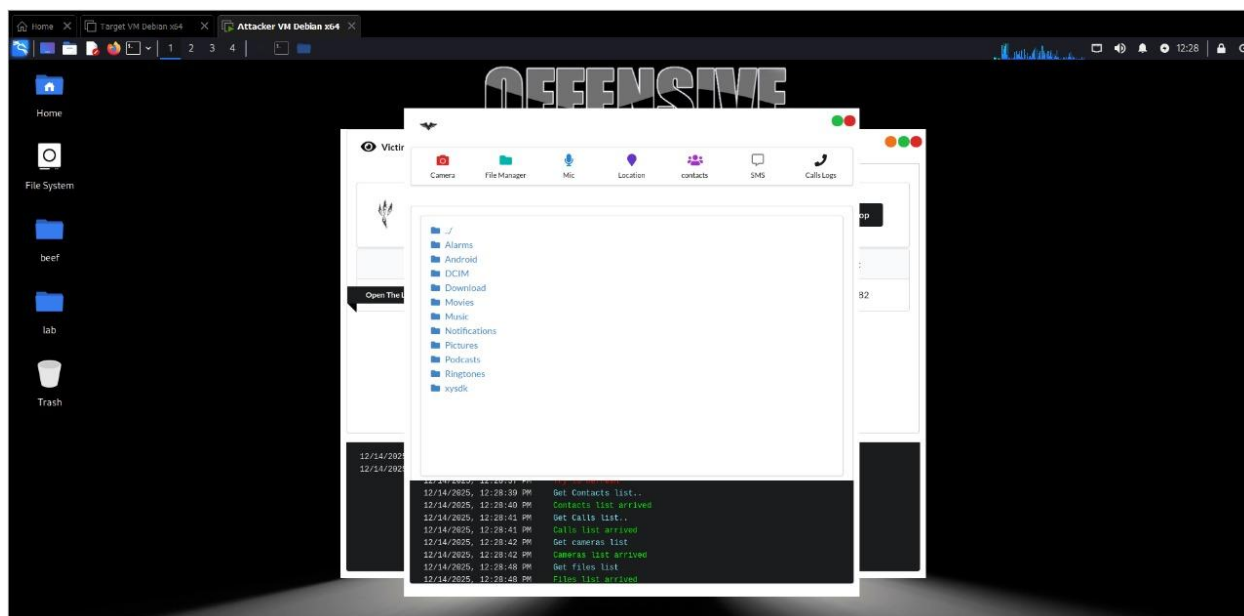
In our case, when AhMyth had created the apk file for Flappy Bird with infused malware, we shared the file to our android virtual machine on Memu play and installed it over there.



After installation was complete, we started listening for devices on AhMyth and then launched the app. When the app was launched, It was shown on AhMyth's interface that a victim has been connected.



There onwards, we were able to access all the files stored in the virtual phone, all call logs, messages, location etc. We could also access the laptop or phone's camera if the virtual machine was given camera access.



As soon as we deleted the app, the victim would get disconnected from AhMyth, which would also be portrayed on its interface. This way we simulated all our attacks on devices using AhMyth.

Safety Measures Against RATs

How RATs Infect Systems:

Most RAT infections happen because of unsafe user actions. Attackers commonly spread RATs through email attachments, fake links, or malicious downloads. Cracked software and fake updates are also popular methods. In some cases, outdated software gives attackers an easy way in. Being aware of these entry points helps users avoid many attacks. Following are some ways to protect devices against RAT attacks:

1. Keeping Software Updated:

Outdated software creates security gaps. Attackers take advantage of these gaps to install RATs. Regular updates fix these problems and reduce the chance of infection. Enabling automatic updates makes this process easier and more reliable.

2. Using Antivirus and Malware Protection:

Antivirus software plays an important role in protection. It detects known threats and blocks suspicious files. Real-time protection helps stop malware before it spreads. Running full system scans from time to time can also help find threats that try to stay hidden.

3. Safe Download Habits:

Many RATs enter systems through unsafe downloads. Users should avoid pirated software and unknown websites. Downloading programs only from official sources lowers risk. Checking file extensions can also prevent accidental installation of harmful files.

4. Email Phishing Awareness:

Email is still one of the easiest ways for attackers to spread RATs. Many people get infected simply by clicking on the wrong link or opening an attachment without thinking twice. Messages that feel rushed, unexpected, or slightly “off” should not be trusted. Attachments from unknown senders and emails asking users to enable macros are especially risky. Taking a few extra seconds to check who sent the email can prevent a lot of trouble.

5. Network and System Protection:

Network security does not have to be complicated. A basic firewall already blocks many unwanted connections. This makes it harder for a RAT to communicate with the attacker. Systems should also avoid running services that are not needed, such as

remote access tools that are never used. The fewer open doors a system has, the safer it becomes.

6. User Behavior and Monitoring:

Using a standard user account instead of an administrator account limits how much control a RAT can gain. Users should also pay attention to warning signs. These include slow performance, unknown programs starting automatically, or unusual network activity.

7. Backup and Recovery:

Regular backups protect important data. Storing backups offline or in secure cloud storage allows recovery if a system becomes compromised. Backups reduce stress and data loss after an attack.

What to Do If a RAT Is Suspected?

If a RAT is suspected, the apk consisting the malware should be immediately deleted, and the system should be disconnected from the internet right away. This limits further damage. A full malware scan should follow. Passwords should be changed using a clean device. In serious cases, reinstalling the operating system provides the safest solution.

Conclusion

Remote Access Trojans are a hidden but serious threat. They rely on ignorance and user mistakes. Basic security habits, regular updates, and awareness greatly reduce risk. With the right precautions, users can protect their systems and data more effectively. AhMyth is used to portray how these RATs can affect us but can also be used to achieve mischievous goals of attackers. This tool is solely to test android vulnerabilities, hence the open source nature and should only be used with this intent.